

MATLAB

matlab.mathworks.com

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LABELS AND ANNOTATIONS

Files

Workspace

EX1.m

ex2.m

ex3.m

EX4.m

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EX6.m

Figure 1

clear; close all; clc;

%% Parameters

Fs = 600e3; % Sampling frequency

t = 0:1/Fs:5e-3; % 5 ms duration

fm = 5e3; % Message frequency (≈ 5 kHz bandwidth)

fc = 20e3; % Carrier frequency 20 kHz

%% Message signal

m = sin(2*pi*fm*t);

%% Carrier signal

c = cos(2*pi*fc*t);

%% Modulated signal (DSB-SC)

s = m .* c;

%% ----- PLOTTING -----

figure;

subplot(4,1,1);

plot(t(1:1000), m(1:1000), 'LineWidth',1.3);

title('Message Signal (5 kHz)');

xlabel('Time (s)'); ylabel('Amplitude');

Command Window

Spectrum Peak Detected at = -15.1 kHz

>>

Message Signal (5 kHz)

Carrier Signal (20 kHz)

Modulated Signal (Message × Carrier)

Spectrum of Modulated Signal

36°C

Partly sunny

Search

ENG IN

14:49

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